

TECHNICAL SPECIFICATION FOR G.I. EARTHING ROD

1.	<u>SCOPE :</u> This specification covers the technical details of G.I. Earthing Rod complete with the necessary fittings.
2	The earthing rod should be 1853 mm (i.e. 25 + 1752 + 76) long fabricated from 20 mm dia M.S. Rod, the bottom of which to be made cut of the same rod in the shape of a cone – 76 mm long and the forged head made out of the same rod with 30 mm (dia) = 25 mm (height). The earthing arrangements should consist of G.I. Bolt/nut and washers. The earthing rod will be as per this office drawing No.S&P/MC/Misc/8. The raw materials, as required for manufacture, shall comply with the relevant latest Indian Standard with all amendments, additions and alternation, for obtaining the required strength.
3	The rod including the head portion should be smoothly and continuously hot dip galvanised as per relevant I.S.S. Other portion i.e. Bolts, nuts and washers should be hot dip galvanised or Zinc electro-plated. For Zinc electro plating manufacturers will have to give undertaking for satisfactory performance in the Test Certificate while offering inspection.
4	No crack should develop and deformation in the top head and/or bending of rod should not be appreciable while the rods will be driven into the ground by the application of heavy intermittent block not less than 7.5 Kg. Hammer and in a manner as usual for driving rods into the grounds.
5	<u>GUARANTEE, TESTING & INSPECTION :</u> Eight copies of test certificates should be furnished along with the offer of inspection. Following tests will be carried out by our inspecting Engineers before acceptance of any materials :
i)	Galvanising tests as per the relevant I.S.S. On 1% of the offered lot upto the maximum of 5 Nos.
ii)	Dimensional check on 1% of the offered lot upto the maximum of 10 Nos.
iii)	Stores test by hammering (7/8 times) the rods into the normal soil by a hammer (not less than 7.5 Kg) on 1% of the offered lot upto the maximum of 2 nos.
6	<u>ACCEPTANCE CRITERIES</u> <u>FOR 1ST SAMPLE</u>
A. (a)	Failure in Galvanising test and dimensional check upto 1/5th will conform the lot for acceptance. But in case of dimensional check failure upto 1/5th will be restricted upto as follows :
i)	1% on total length
ii)	7½% on other dimension except 15 mm, 25 mm & 40 mm of small forged portion.
iii)	No failure will be accepted in 15 mm, 25 mm, & 40 mm of small forged portion. If the failure is beyond the above tolerance, re-sampling may be made as per clause 6

B.	If the failure in each of the above tests are more than $1/5^{\text{th}}$ but less than or equal to $2/5^{\text{th}}$ resampling in the same mannar will be made for the test(s) in which failed.
C.	However, if the failure in any of the above tests is greater than $2/5^{\text{th}}$ the whole lot will be rejected.
D.	<u>For 2nd Sample :</u> In case of failure as (a) above the lot will conform for acceptance. However, if the failure in any of the above tests is greater than $1/5^{\text{th}}$ the whole lot will be rejected. No resampling will be made in case of failure in stress tests by hammering. Any defect found during Inspection/Testing of the Earthing rod or after delivery in the site in respect of Galvanising, welding and in the quality of the associated materials shall make the whole lot liable for rejection. The rejected materials shall be taken back by the supplier at his own cost from the place of delivery within ten days from the date of intimation in this respect. Earthing Rods with all fittings mounted on them shall be supplied in boundles of ten. <u>TOLERANCE :</u> - $\frac{1}{2}\%$ on total length., - 5% on other dimensions. - 10% on 15 mm, 25 mm, & 40 mm of small forged portion. Any tolerance on the positive side will be accepted.
9	<u>MARKING :</u> Manufacturers identification marking is to be punched on the Head of the Earth Rod.

1	<u>TECHNICAL SPECIFICATION FOR GALVANISED STAY SET :</u>		
	<u>SCOPE :</u> This specification covers the details of the Galvanised stay set both in HT & LT complete with stay rod, stay plate, Bow, Cross Head, Ratchet Nut and a Thimble		
2	The sizes of stay rods are stated below, Linensions for other parts like bow plate etc. are not stated in details here and shall be as per this office drawing No.S7P/MC/MISC/7 and S&P/MC/Misc/6 for H.T. & L.T. Respectively.		
	<u>Length</u>	<u>Dia</u>	<u>Length of the threaded portion</u>
i)	1830 mm	20 mm	300 mm (H.T.)
ii)	1680 mm	16 mm	300 mm (L.T.)
	The raw materials as required for manufacture shall comply with the relevant latest Indian Standard with all amendments, additions and alternation, for obtaining the required strength.		
3	<u>MECHANICAL PARTICULARS :</u> The complete 1830 mm & 1680 mm long stay sets, should with stand minimum breaking loads of 7900 Kg. & respectively.		
4.1	<u>GENERAL REQUIREMENT :</u> The rods, cylindrical portion of the bow, plates, nuts and thimble shall be of steel to comply with the requirements as stated in class-2 of this specification.		
4.2	The cross-head of the bow shall be made of sound, strong iron casting.		
4.3	The rods shall be well forged and free from flaws and other defects and the heads shall not fail when the rods are tested to fracture at their full strength by tensile stress. The rods shall have square neck.		
4.4	The sides of each bow shall be well revetted into the cross head and shall not come out from the cross head when the bow is tested to fracture by tensil stress.		
4.5	The ratchet nuts and ratchet face of cross heads shall be well from to match each other/		
4.6	The screw thread design of the stay rod and nut should be as follows :		
	M 16 x 2 -7H/8g for 16 mm dia stay rods and Y 20 x 2.5-7H/8g for 20 mm rod and the sizes of the threads should be as per IS: 4218 (Part-VI) 1967 (tables for coarse pinch series). The screw threads of rods and nuts shall be properly lubricates at the time of supply.		
4.7	The stay plate shall be freely out off and punched and shall be free from cracks after punching.		
4.8	The rods, cross heads, bows, ratchet nuts, thimbles and plates shall be smoothly and continuously hot dip galvanised. The galvanishing shall be heavy. The screw threads of the rods, ratchest nuts shall be out after galvanishing. The nuts shall be well finished before galvanishing		
4.9	The Thimbles shall be made from 2.64 mm (12 SWG) thick M.S. Fit. The size of the thimbles shall be follows :		

	<table><tr><td></td><td><u>H. T.</u></td><td><u>L.T.</u></td></tr><tr><td>Length :</td><td>70 mm</td><td>65 mm</td></tr><tr><td>Breadth :</td><td>47 mm</td><td>45 mm</td></tr><tr><td>Width :</td><td>20 mm</td><td>16 mm</td></tr></table>		<u>H. T.</u>	<u>L.T.</u>	Length :	70 mm	65 mm	Breadth :	47 mm	45 mm	Width :	20 mm	16 mm
	<u>H. T.</u>	<u>L.T.</u>											
Length :	70 mm	65 mm											
Breadth :	47 mm	45 mm											
Width :	20 mm	16 mm											
5	<u>TEST & TEST CERTIFICATE :</u>												
5.1	The quality of galvanising shall be determined by tests as given in the relevant ISS with all amendments and alternations.												
5.2	The finished and complete stay seats when subjected in tensile stress as per relevant IS shall comply with the minimum breaking load as stated in clause – 3 of this specification.												
5.3	Eight copies of test certificate shall have to be furnished along with the offer of inspection for our approval.												
5.5	In case, the complete stay sets are found, even subsequent to delivery failing to comply with relevant tests,s the whole lots will be liable for rejection. The rejected materials shall have to be taken back by the supplier at his own cost from the purchaser's store within seven days from the date of receipt of intimation in this respect.												
6	<u>INSPECTION :</u> The purchaser reserves the right to depute representative to witness the production and testing at the Works of the manufacturer. The manufacturer shall afford all reasonable facilities without charge to the inspector representing the purchaser to satisfy him that the material are being furnished in accordance with the specification. The Purchaser also reserves the right to reject the raw materials or the finished products if the same are fund to be not conforming with the requiremeant of this specification. The following tests will be carried out by our inspecting Engineers before acceptance of any materials. 1) Tensile strength test for a breaking load on 1% of the offered lot upto a maximum of 5 nos. 2) Galvanizing test as per the relevant ISS will be carried out on 1% of the offered lot upto a maximum of 5 nos. 3) Dimensional check up on 1% of the offered lot upto a maximum of 10 nos.												
7	<u>ACCEPTANCE CRITERIA :</u>												
A.	<u>For 1st sample :</u> a) Failure upto 1/5 th of individual components in each of the above tests will conform the lot for acceptance, But in case of dimentional check, failure ujpt 1/5 th will be restricted upto as follows: i) 1% on total length of the stay rod. ii) 7½% on individual portion except rod length. b) If the filure in any of the individual components in each of the above tests is more than 1/5 th but less than or equal to 2/5 resampling in the same manner will be made for the test(s) in which failed. c) However if the failure of the individual components in any of the above tests is greater than 2/5 th . The whole lot will be retested.												

B.	<u>FOR 2ND SAMPLES :</u> In case of failure as (a) above the lot will confirm for acceptance. However if the failure of the individual components in each of the above tests is greather than 1/5th the whole lot will be rejected.
8	<u>MARKING :</u> Manufacturer's Identification maki8ng are to be embossed on the casting portion of the bow and to be punched on the Head of the Stay Rod.
9	<u>PACKING :</u> The stay plates shall be packed separately in an approved manned and shall contain not more than 20 plates in one packing. The bow with cross head, thimble and ratchet nut shall be fitted in the stay rod and bundled together so as to contain not more than 10 such sets in any boundle. The threaded portion of the stay sets be properly protected from damage.
10	<u>TOLLERING :</u> The following tolerance will be allowed in case of stay set both H.T. & L.T. a) – 5% tolerance on individual portion except rod length. b) – ½ % tolerance on total length of the stay rod. Any tolerance on positive side will be acceptable.
11	<u>DRAWING :</u> The drawings enclosed with the specification will be applicable. No further submission & approval of drawings will be required.